

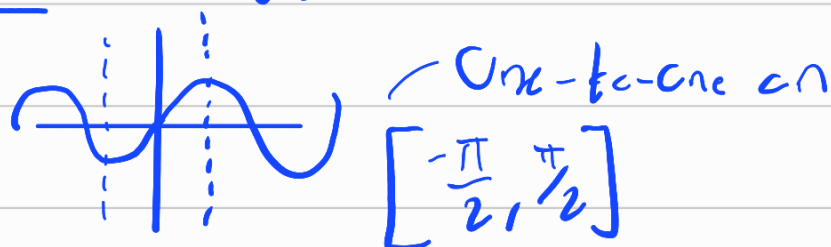
Given a function $f(x)=y$, we often want to know for what y , does $f(x)=y$. We can make another function f^{-1} s.t. $f^{-1}(y)=x$. This is called an inverse function.

The question is now, what function can you invert?

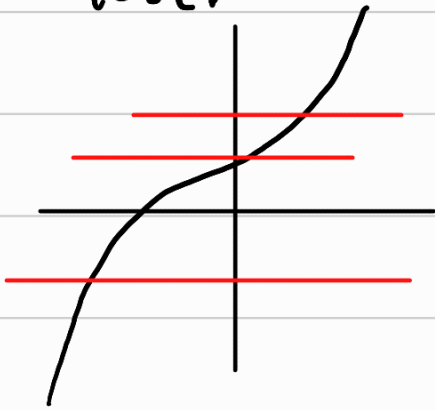
Def: A one-to-one function is a function that never takes on the same value twice. That is, different inputs map to different outputs. i.e. when $x_1 \neq x_2$, $f(x_1) \neq f(x_2)$

* Functions are invertible only on the domain in which they are one-to-one.

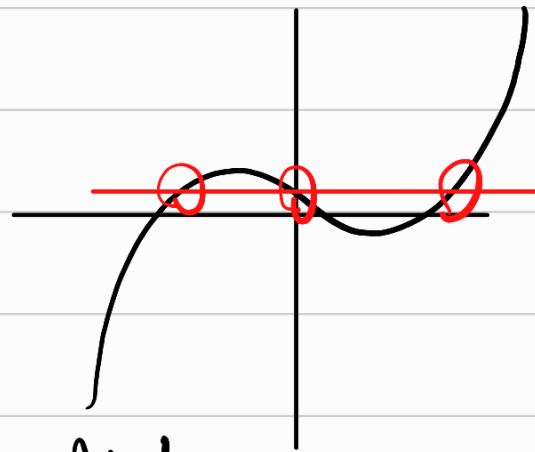
Ex: $\sin(x)$



A one-to-one function must pass horizontal line test.



One-to-One



Not One-to-One

Def: we denote f^{-1} to be the inverse to f .

If f has domain A and Range B
 f^{-1} has domain B and Range A .

and if $f(x) = y \Leftrightarrow f^{-1}(y) = x$

$$f^{-1}(f(x)) = x, \text{ for all } x \in A$$

$$f(f^{-1}(x)) = x, \text{ for all } x \in B$$

finding inverse functions

1. write $y = f(x)$
2. Solve for x if possible (in terms of y)
3. Change $x \rightarrow y$ and $y \rightarrow x$. then you have $y = f^{-1}(x)$

Ex:

$$f(x) = x^3 + 2$$

$$y = x^3 + 2$$

$$y - 2 = x^3$$

$$\sqrt[3]{y-2} = x$$

$$\rightarrow f^{-1}(x) = \sqrt[3]{x-2}$$

Note:

always increasing
or always decreasing

A Strictly Monotone function is always
one-to-one

Derivatives of inverse function

Thm: If a function f is continuous and one-to-one then so too is f^{-1} .

Thm: If f is one-to-one and differentiable with inverse f^{-1} and $f'(f^{-1}(a)) \neq 0$, then

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

PP/ Let $f(b)=a$, $f(y)=x$ then
 $f^{-1}(a)=b$, $f^{-1}(x)=y$.

So

$$(f^{-1})'(a) = \lim_{x \rightarrow a} \frac{f^{-1}(x) - f^{-1}(a)}{x - a}$$

$$= \lim_{y \rightarrow b} \frac{y - b}{f(y) - f(b)}$$

$$= \lim_{y \rightarrow b} \frac{1}{\frac{f(y) - f(b)}{y - b}}$$

$$= \frac{1}{f'(b)} = \frac{1}{f'(f^{-1}(a))}$$

Ex:

$$f(x) = x^2, \text{ find } (f^{-1})'(1)$$

$$x^2 = 1 \text{ if } x = 1, \text{ so } f^{-1}(1) = 1$$

$$\text{then } (f^{-1})' = \frac{1}{f'(f^{-1}(1))} = \frac{1}{f'(1)} = \frac{1}{2(1)} \\ = \frac{1}{2}.$$